DIE INSEL

BUSINESS INTELLIGENCE & OPERATIONS ANALYTICS

Turning everyday business records into practical insights for better sales, product and inventory decisions.

THE BUSINESS OBJECTIVE

To transform everyday business records into organised, meaningful information that helps management understand performance, monitor operations and make more informed decisions.

FOUR AREAS OF BUSINESS INTELLIGENCE

01. *BUSINESS PERFORMANCE*
-Revenue and overall performance monitoring

02. *SALES ANALYSIS*
-Sales patterns and transaction activity

03. *PRODUCT INTELLIGENCE*
-Product performance and revenue contribution

04. *INVENTORY MONITORING*
-Recorded stock levels and stock movements

Business Analytics Demonstration | Mombasa, Kenya

UNDERSTANDING THE BUSINESS

Making everyday business information more useful for management decisions.

SALES

Understand what is selling, when sales are highest and how revenue changes over time.

PRODUCTS

Identify popular products, stronger-performing categories and their contribution to revenue.

INVENTORY

Monitor recorded stock levels, stock movements and potential stock shortages.

Business records become more valuable when they help answer practical questions:

- Which products contribute most to revenue?
- When does the business experience its busiest sales periods?
- Which products require closer stock monitoring?
- Where can management investigate opportunities to improve performance?

The objective is to move from simply recording business activity to understanding what the information reveals.

FROM BUSINESS RECORDS TO ACTIONABLE INSIGHTS

A STRUCTURED PROCESS THAT TURNS EVERYDAY SALES, PRODUCT AND STOCK INFORMATION INTO MEANINGFUL BUSINESS INTELLIGENCE.

1.BUSINESS RECORDS

Sales transactions, product information, inventory records and stock movements.

2.DATA PREPARATION

Python and Pandas help organise, clean and validate the records for analysis.

3.DATABASE & ANALYSIS

MySQL stores the structured information, while SQL queries prepare useful business summaries.

4.MANAGEMENT DASHBOARD

Visual reports help management monitor performance, explore trends and identify areas needing attention.

WHAT DOES THIS MEAN FOR THE BUSINESS?

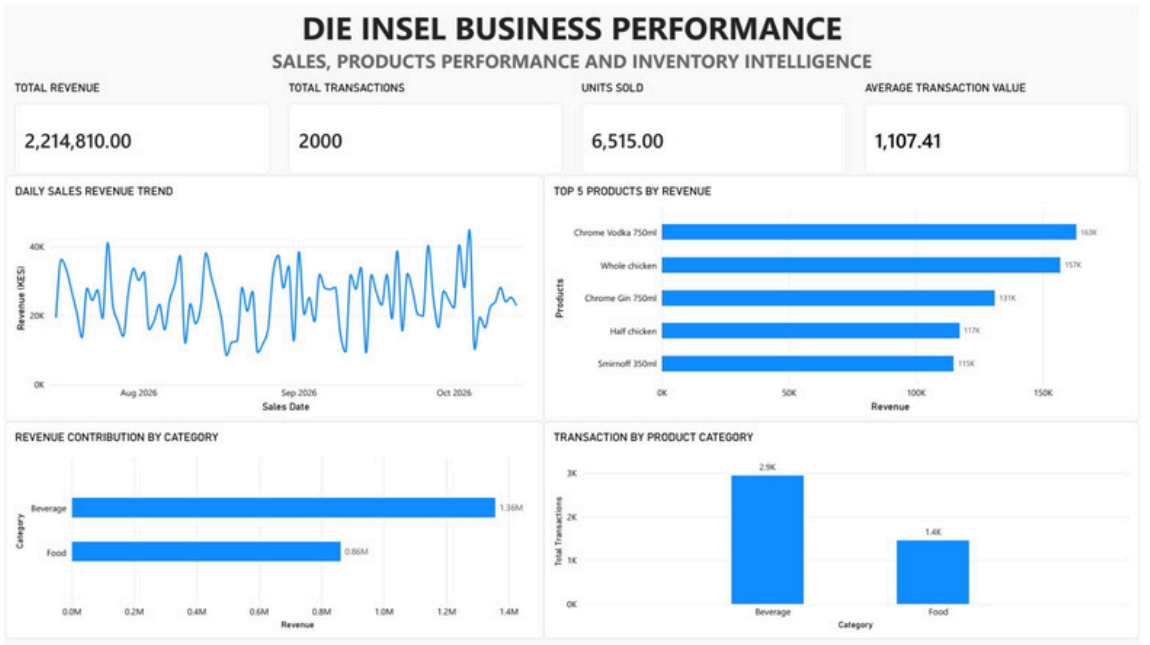
-I*nstead of reviewing separate records manually, management can view organised information in one reporting environment.*

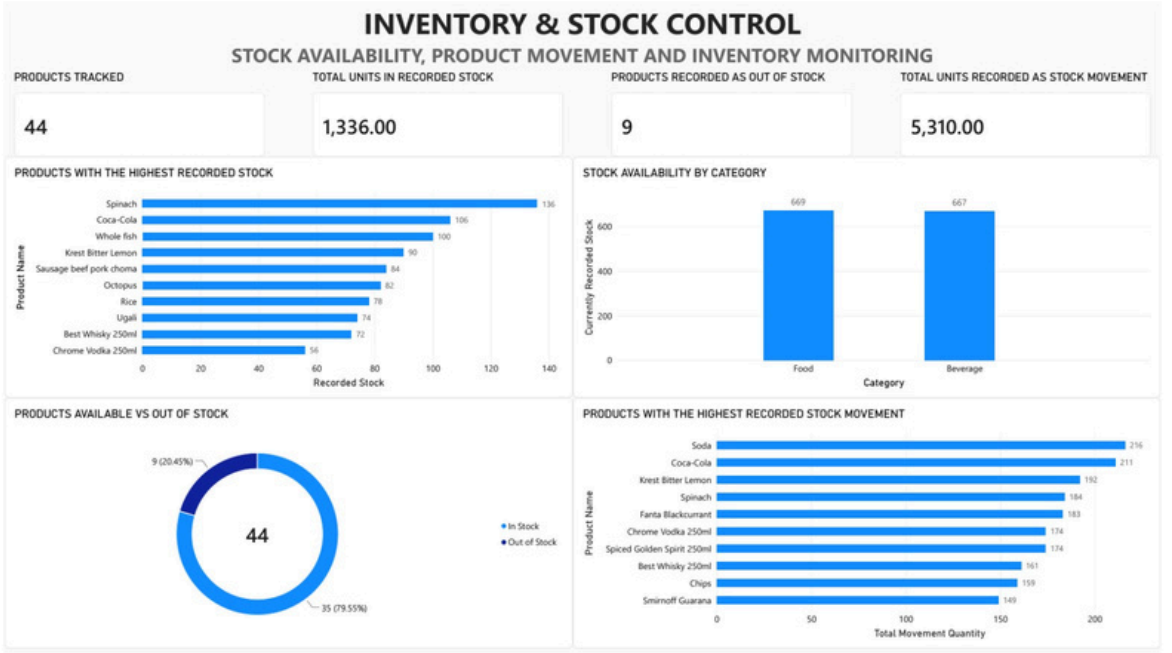
-The dashboard brings together sales performance, customer activity patterns, product contribution and recorded inventory information to support more informed operational decisions.

DATA SOURCES → PYTHON / PANDAS → MYSQL / SQL → POWER BI

THE DASHBOARD: TURNING DATA INTO BUSINESS INSIGHTS

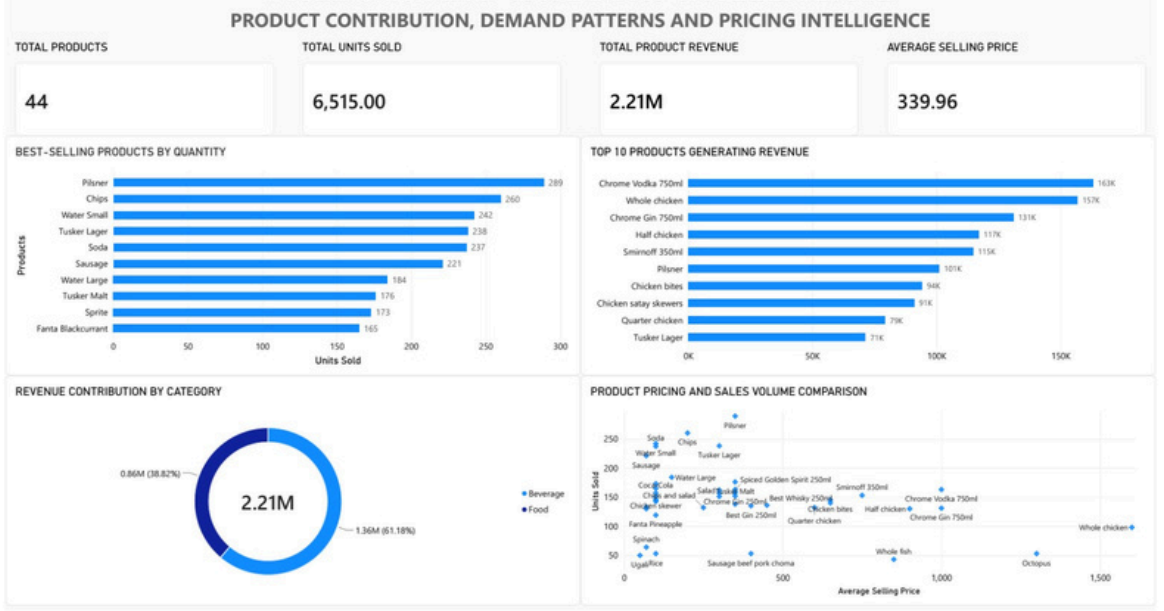
FOUR AREAS OF ANALYSIS DESIGNED TO HELP MANAGEMENT UNDERSTAND BUSINESS PERFORMANCE AND MONITOR DAILY OPERATIONS.





Understand overall revenue, transaction activity, units sold and average customer spend.

Explore sales patterns, busy periods, daily activity & changes in average transaction value.





Identify popular products, revenue contribution and differences in product performance.

Review recorded stock levels, stock movements and products that may require attention.

NEXT STEP

-Review the existing business records, confirm data availability and reporting requirements, and determine how the solution can be adapted to the business's actual operations.